Vector potentials in three-dimensional non-smooth domains: a Lean blueprint

Formalization of C. Amrouche, C. Bernardi, M. Dauge and V. Girault (1998)

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Chapter 1

Overview

This blueprint describes a Lean 4 formalization of the two theorems on canonical vector potentials of

> C. Amrouche, C. Bernardi, M. Dauge, V. Girault, *Vector potentials in three-dimensional non-smooth domains*, Math. Methods Appl. Sci. **21** (1998), no. 9, 823–864.

Let $\Omega \subset \mathbb{R}^3$ be a bounded connected Lipschitz domain whose boundary has the connected components $\Gamma_0, \dots, \Gamma_I$ (Γ_0 the outer one), and let $\Sigma_1, \dots, \Sigma_J$ be an admissible system of cuts, so that $\Omega^\circ = \Omega \setminus (\Sigma_1 \cup \cdots \cup \Sigma_J)$ is simply connected. For $u \in L^2(\Omega)^3$:

- **Theorem 3.12.** $\operatorname{div} u = 0$ and u has zero flux through every Γ_i if and only if there is exactly one ψ with $\operatorname{curl} \psi = u$, $\operatorname{div} \psi = 0$, $\psi \cdot n = 0$ on $\partial\Omega$, and zero flux through every cut Σ_j (the *tangential* vector potential).
- **Theorem 3.17.** $\operatorname{div} u = 0$, $u \cdot n = 0$ on $\partial\Omega$ and u has zero flux through every cut if and only if there is exactly one ψ with $\operatorname{curl} \psi = u$, $\operatorname{div} \psi = 0$, $\psi \times n = 0$ on $\partial\Omega$, and zero flux through every Γ_i (the *normal* vector potential).

The Lean files.

- `Showcase.lean` imports only Mathlib. It defines the setting and states both theorems in the notation of the paper ($\operatorname{div} u = 0$, $\operatorname{curl} \psi = u$, $\psi \cdot n = 0$, $\psi \times n = 0$, $F_i(u) = 0$, $C_j(\psi) = 0$), with `sorry`.
- `Showcase_WithProofs.lean` proves the same theorems from the library `LeanCode`.
- `comparator.json` configures `lake comparator`. It checks that the proofs establish exactly the statements of `Showcase.lean` and use only the axioms `propext`, `Classical.choice` and `Quot.sound`.
- The library `LeanCode` contains about 8 400 lines of proofs, organized as in the chapters below. `LeanCode/Definitions.lean` repeats the definitions of `Showcase.lean` verbatim.
- `blueprint/DetailedProofs.pdf` contains detailed mathematical proofs of all principal results of the paper. Its arguments for the two theorems formalized here partly differ from the route of the Lean proof described below.

The route. The proof has two layers. An abstract Hilbert-space argument (Chapter 5) derives both theorems from seven analytic facts, collected in Definition 5.1. The other chapters prove these facts for every admissible geometric datum: the cuts are null and carry unit-jump lifts (Chapter 6); curl-free, divergence-free fields are smooth (Chapter 7) and, on the simply connected cut domain, gradients (Chapter 8); and divergence-free fields with zero boundary fluxes extend to solenoidal fields on $\mathbb{R}^3$, which are curls of an explicit line-integral potential (Chapter 9).

Chapter 2

Results from Mathlib

The formalization builds on Mathlib. The following definitions and results of Mathlib are its main ingredients; in the dependency graph they are grey, and their edges show where they enter the proofs.

Definition 2.1 (The Hilbert space L^2). [Lean: `MeasureTheory.Lp`, `MeasureTheory.L2.innerProductSpace`, `MeasureTheory.Lp.instCompleteSpace`] For a measure μ, the space $L^2(\mu)$ of square-integrable functions, taken modulo equality almost everywhere, with its inner product; it is complete.

Definition 2.2 (Test functions). [Lean: `TestFunction`] The space $C_c^\infty(\Omega)$ of smooth functions with compact support in an open set Ω.

Theorem 2.3 (Orthogonal projection). [Lean: `Submodule.starProjection`, `Submodule.orthogonal_orthogonal_eq_closure`] *A complete subspace K of an inner product space has an orthogonal projection P_K, with $x - P_K x \in K^\perp$; moreover $(K^\perp)^\perp = \overline{K}$.*

Theorem 2.4 (Riesz representation theorem). [Lean: `InnerProductSpace.toDual`] *Every bounded linear functional on a Hilbert space is of the form $x \mapsto (y, x)$ for a unique y.*

Theorem 2.5 (Closed graph theorem). [Lean: `LinearMap.continuous_of_isClosed_graph`] *A linear map between Banach spaces whose graph is closed is continuous.*

Theorem 2.6 (Hahn–Banach theorem). [Lean: `exists_extension_norm_eq`] *A bounded linear functional on a subspace of a normed space extends to the whole space with the same norm.*

Theorem 2.7 (Rademacher's theorem). [Lean: `LipschitzWith.ae_differentiableAt`, `LipschitzWith.integral_lineDeriv_mul_eq`] *A Lipschitz function on $\mathbb{R}^n$ is differentiable almost everywhere, and it can be integrated by parts against Lipschitz functions with compact support along every direction.*

Theorem 2.8 (Poincaré lemma on convex sets). [Lean: `Convex.exists_forall_hasFDerivAt_of_fderiv_symmetric`] *On an open convex set, a differentiable field whose derivative is symmetric is the gradient of a function.*

Theorem 2.9 (Lifting through covering maps). [Lean: `IsCoveringMap.existsUnique_continuousMap_lifts`] *A continuous map from a simply connected, locally path-connected space lifts through a covering map, uniquely once the lift of one point is chosen.*

Theorem 2.10 (Fubini–Tonelli theorem). [Lean: `MeasureTheory.integral_prod`, `MeasureTheory.lintegral_prod`, `MeasureTheory.integral_integral_swap`] *Integrals over a product of σ-finite measure spaces are iterated integrals, in either order.*

Theorem 2.11 (Dominated convergence). [Lean: `MeasureTheory.tendsto_integral_of_dominated_convergence`, `hasFDerivAt_integral_of_dominated_of_fderiv_le`] *Dominated convergence theorem, and differentiation under the integral sign under a dominating bound.*

Theorem 2.12 (Bump functions and convolution). [Lean: `ContDiffBump, HasCompactSupport.contDiff_convolution_left`] *There are smooth bump functions with compact support, and the convolution of a locally integrable function with a smooth compactly supported kernel is smooth.*

Theorem 2.13 (Smooth partitions of unity). [Lean: `SmoothPartitionOfUnity.exists_isSubordinate`] *Every open cover of a closed subset of $\mathbb{R}^n$ has a subordinate smooth partition of unity.*

Chapter 3

Setting and definitions

All definitions are in `Showcase.lean`, namespace `VP`. They are repeated verbatim in `LeanCode/Definitions.lean`.

Definition 3.1 (Fields, weak derivatives and Sobolev spaces)**.** [Lean: `VP.Domain, VP.Scalar, VP.Field, VP.ScalarTest, VP.VectorTest, VP.HasWeakGradient, VP.HasWeakDivergence, VP.HasWeakCurl, VP.HasWeakJacobian, VP.H1Scalar, VP.H1Vector, VP.DivFree`] Let $\Omega \subset \mathbb{R}^n$ be open. $L^2(\Omega)$ and $L^2(\Omega)^n$ are the square-integrable functions and vector fields, as classes modulo equality almost everywhere; $C_c^\infty(\Omega)$ and $C_c^\infty(\Omega)^3$ are the smooth compactly supported test functions and test fields. For $v, c \in L^2(\Omega)^3$ and $d \in L^2(\Omega)$:

- $\operatorname{div} v = d$ if $\int_\Omega v \cdot \nabla\varphi = -\int_\Omega d\,\varphi$ for all $\varphi \in C_c^\infty(\Omega)$; v is *divergence free* if $\operatorname{div} v = 0$;
- $\operatorname{curl} v = c$ if $\int_\Omega v \cdot \operatorname{curl}\varphi = \int_\Omega c \cdot \varphi$ for all $\varphi \in C_c^\infty(\Omega)^3$;
- an element of $H^1(\Omega)$ is a pair of $q \in L^2(\Omega)$ and its weak gradient $\nabla q \in L^2(\Omega)^n$; an element of $H^1(\Omega)^3$ is a pair of a field and its weak Jacobian in L^2.

Definition 3.2 (Zero normal and tangential traces)**.** [Lean: `VP.ZeroNormalTrace, VP.ZeroTangentialTrace`] Let $v, c \in L^2(\Omega)^3$ and $d \in L^2(\Omega)$.

- For a field with $\operatorname{div} v = d$, $v \cdot n = 0$ *on* $\partial\Omega$ means $\int_\Omega v \cdot \nabla q + \int_\Omega d\,q = 0$ for all $q \in H^1(\Omega)$.
- For a field with $\operatorname{curl} v = c$, $v \times n = 0$ *on* $\partial\Omega$ means $\int_\Omega v \cdot \operatorname{curl} w - \int_\Omega c \cdot w = 0$ for all $w \in H^1(\Omega)^3$.

These are the paper's weak definitions by Green's formulas; no trace operator is needed.

Definition 3.3 (Admissible geometric data)**.** [Lean: `VP.GraphPatch, VP.LipschitzBoundary, VP.BoundaryComponents, VP.CutSidePatch, VP.CutSystem, VP.LipschitzData, VP.GeometricData, VP.cutDomain`] A *graph patch* is an open box, in coordinates given by an affine isometry of $\mathbb{R}^3$, together with a Lipschitz function of the first two coordinates; it splits the box into the regions above and below the graph. A set has *Lipschitz boundary* if every boundary point lies in a patch in which the set is the region above the graph and its boundary is the graph. Geometric data of an open set $\Omega \subset \mathbb{R}^3$ consist of:

- the facts that Ω is bounded, connected and has Lipschitz boundary;
- the boundary components $\Gamma_0, \ldots, \Gamma_I$: the connected components of $\partial\Omega$, pairwise disjoint and covering $\partial\Omega$, where Γ_0 borders the unbounded component of $\mathbb{R}^3 \setminus \overline{\Omega}$;
- a cutting system $\Sigma_1, \ldots, \Sigma_J$ (numbered from 0 in Lean): connected subsets of Ω with rims on $\partial\Omega$ and pairwise disjoint closures. Each cut is covered by finitely many balls in which $\Omega \setminus \Sigma_j$ splits into a connected Lipschitz *minus* side and a connected Lipschitz *plus* side, the cut being a smooth graph between them with coherent orientation. The cut domain $\Omega^\circ = \Omega \setminus \bigcup_j \Sigma_j$ is open, simply connected and pseudo-Lipschitz (near each boundary point it consists of one or two Lipschitz pieces).

No extension, Poincaré, trace, closed-range or potential theorem is part of these data.

Definition 3.4 (Boundary fluxes and cut fluxes). [Lean: `VP.IsBoundaryCutoff`, `VP.ZeroBoundaryFluxOn`, `VP.ZeroBoundaryFlux`, `VP.HasConstantJump`, `VP.HasUnitCutJump`, `VP.ZeroCutFlux`] Let $v \in L^2(\Omega)^3$ with $\operatorname{div} v = d$.

- A *cutoff* for Γ_i is a smooth compactly supported η on $\mathbb{R}^3$ that is 1 near Γ_i and 0 near the other components. $F_i(v) = 0$ means $\int_\Omega v \cdot \nabla\eta + \int_\Omega d\,\eta = 0$ for every cutoff η of Γ_i, for all $0 \le i \le I$.
- $q \in H^1(\Omega^\circ)$ *jumps by* c *across* Σ_j if, in each ball of the patches of Σ_j, the function equal to $q - c$ on the minus side and to q on the plus side is the restriction of an H^1 function. It has *unit jump across* Σ_j if it jumps by 1 across Σ_j and by 0 across the other cuts (the paper's lifts r_j).
- $C_j(v) = 0$ means $\int_{\Omega^\circ} v \cdot \nabla r + \int_{\Omega^\circ} d\,r = 0$ for every r with unit jump across Σ_j, for all j.

The conditions are imposed for every admissible cutoff and lift, so no choice is involved.

Definition 3.5 (Tangential and normal potentials). [Lean: `VP.IsTangentialPotential`, `VP.IsNormalPotential`] Let $u, \psi \in L^2(\Omega)^3$. Then ψ is a *tangential potential* of u if $\operatorname{curl}\psi = u$, $\operatorname{div}\psi = 0$, $\psi \cdot n = 0$ and $C_j(\psi) = 0$ for all j, and a *normal potential* of u if $\operatorname{curl}\psi = u$, $\operatorname{div}\psi = 0$, $\psi \times n = 0$ and $F_i(\psi) = 0$ for all i. In `Showcase.lean` the theorems are written with the scoped notation of the paper (namespace `VP.Paper`), which abbreviates exactly these conditions; the file checks each abbreviation by `rfl`.

Lemma 3.6 (The flux conditions are not vacuous). [Lean: `VP.exists_boundaryCutoff`, `VP.liftH1_unitJump`] *Every boundary component Γ_i has a cutoff, and every cut Σ_j has a function in $H^1(\Omega^\circ)$ with unit jump across Σ_j.*

Proof. The components are compact and pairwise disjoint, so a smooth cutoff equal to 1 near Γ_i and supported away from the other components exists. The lifts are those of Lemma 6.2. □

Chapter 4

The main theorems

Theorem 4.1 (Theorem 3.12: tangential vector potentials)**.** [Lean: `VP.tangentialPotential_iff`, `VP.theorem_3_12`] *Let Ω carry admissible geometric data and let $u \in L^2(\Omega)^3$. Then $\operatorname{div} u = 0$ and $F_i(u) = 0$ for $0 \le i \le I$ if and only if there is exactly one $\psi \in L^2(\Omega)^3$ with*

$$\operatorname{curl} \psi = u, \qquad \operatorname{div} \psi = 0, \qquad \psi \cdot n = 0, \qquad C_j(\psi) = 0 \ (1 \le j \le J).$$

Proof. Theorem 5.4, applied with the analytic inputs of Theorem 5.8. □

Theorem 4.2 (Theorem 3.17: normal vector potentials)**.** [Lean: `VP.normalPotential_iff`, `VP.theorem_3_17`] *Let Ω carry admissible geometric data and let $u \in L^2(\Omega)^3$. Then $\operatorname{div} u = 0$, $u \cdot n = 0$ and $C_j(u) = 0$ for $1 \le j \le J$ if and only if there is exactly one $\psi \in L^2(\Omega)^3$ with*

$$\operatorname{curl} \psi = u, \qquad \operatorname{div} \psi = 0, \qquad \psi \times n = 0, \qquad F_i(\psi) = 0 \ (0 \le i \le I).$$

Proof. Theorem 5.7, applied with the analytic inputs of Theorem 5.8. □

Chapter 5

The abstract assembly

Definition 5.1 (Analytic inputs). [Lean: `VP.AnalyticInputs`] The *analytic inputs* for geometric data G of Ω are the following statements.

1. The cuts are Lebesgue null.
2. If $h \in L^2(\Omega)^3$ is curl free and divergence free, and $v \in L^2(\Omega)^3$ satisfies $v \cdot n = 0$ and $C_j(v) = 0$ for all j, then $(v, h)_{L^2(\Omega)^3} = 0$.
3. Every divergence-free u with $F_i(u) = 0$ for all i is a curl.
4. A divergence-free ψ with $\psi \times n = 0$ (for $\operatorname{curl} \psi = 0$) and $F_i(\psi) = 0$ for all i vanishes.
5. If $\psi \times n = 0$ for a field ψ with $\operatorname{curl} \psi = u$, then $u \cdot n = 0$.
6. The gradient of a function with unit jump across a cut, extended across the null cuts, is curl free on Ω.
7. If $\psi \times n = 0$ for a field ψ with $\operatorname{curl} \psi = u$, and $u \cdot n = 0$, then $C_j(u) = 0$ for all j.

Lemma 5.2 (Necessary conditions). [Lean: `VP.divFree_of_hasWeakCurl`, `VP.zeroBoundaryFlux_of_hasWeakCurl`, `VP.hasWeakCurl_gradient`, `VP.divFree_of_zeroNormalTrace`] *A weak curl is divergence free and has zero flux through every boundary component. Gradients of $H^1(\Omega)$ functions are curl free. A field with $v \cdot n = 0$ (for $\operatorname{div} v = 0$) is divergence free.*

Proof. $\operatorname{curl} \nabla \varphi = 0$ and $\operatorname{div} \operatorname{curl} \varphi = 0$ for smooth test functions. The gradient of a cutoff of Γ_i vanishes near $\partial\Omega$, so it is the gradient of a test function on Ω. The last statement follows by testing $v \cdot n = 0$ with $q = \varphi \in C_c^\infty(\Omega)$. □

Lemma 5.3 (The gauge space). [Lean: `VP.gaugeFree`, `VP.mem_gaugeFree`, `VP.isTangentialPotential_iff`, `VP.AnalyticInputs.gauge_le_curlKernel`] *Assume the cuts are null, and let $K \subset L^2(\Omega)^3$ be the orthogonal complement of the gradients of $H^1(\Omega)$ functions and of the (extended) gradients of functions with unit jump across a cut. Then $v \in K$ if and only if $v \cdot n = 0$ and $C_j(v) = 0$ for all j, and ψ is a tangential potential of u if and only if $\operatorname{curl} \psi = u$ and $\psi \in K$. Under the analytic inputs, every element of $K^\perp$ is curl free.*

Proof. The first two statements unfold the definitions; divergence freeness follows from Lemma 5.2. $K^\perp$ is the closed span of the generating gradients, which are curl free by Lemma 5.2 and input 6; the curl kernel is closed. □

Theorem 5.4 (Assembly of Theorem 3.12). [Lean: `VP.tangentialPotential_iff_of`] *If the analytic inputs hold, then $\operatorname{div} u = 0$ and $F_i(u) = 0$ for all i if and only if u has exactly one tangential potential.*

Proof. By input 3 there is ψ_0 with $\operatorname{curl}\psi_0 = u$. Let ψ be the orthogonal projection of ψ_0 onto K. Then $\psi_0 - \psi \in K^\perp$ is curl free (Lemma 5.3), so $\operatorname{curl}\psi = u$ and ψ is a tangential potential. If χ is another one, $\chi - \psi$ is curl free and lies in K; hence it is divergence free with $(\chi - \psi)\cdot n = 0$ and zero cut fluxes, and input 2 with $v = h = \chi - \psi$ gives $\|\chi - \psi\|^2 = 0$. The converse is Lemma 5.2. □

Lemma 5.5 (The canonical tangential inverse). [Lean: `VP.fluxFree, VP.tangentialInverse, VP.tangentialInverse_continuous, VP.tangentialInverseL`] *The divergence-free fields with zero boundary fluxes form a closed subspace of $L^2(\Omega)^3$. On it, the map sending u to its tangential potential is linear and continuous.*

Proof. Linearity follows from uniqueness. The graph of the map is the set of pairs (u, ψ) with $\operatorname{curl}\psi = u$ and $\psi \in K$, which is closed; the closed graph theorem gives continuity. □

Lemma 5.6 (Orthogonality to the curl kernel). [Lean: `VP.AnalyticInputs.orth_curlKernel`] *Assume the analytic inputs. If $u \cdot n = 0$, $C_j(u) = 0$ for all j, and w is curl free, then $(u, w)_{L^2(\Omega)^3} = 0$.*

Proof. Write $w = w_1 + (w - w_1)$ with w_1 the projection onto the orthogonal complement of the gradients of $H^1(\Omega)$ functions. Then $w - w_1$ is a limit of such gradients, orthogonal to u because $u \cdot n = 0$; and w_1 is curl free with $w_1 \cdot n = 0$, hence divergence free, so $(u, w_1) = 0$ by input 2. □

Theorem 5.7 (Assembly of Theorem 3.17). [Lean: `VP.normalPotential_iff_of`] *If the analytic inputs hold, then $\operatorname{div} u = 0$, $u \cdot n = 0$ and $C_j(u) = 0$ for all j if and only if u has exactly one normal potential.*

Proof. Let T be the canonical tangential inverse of Lemma 5.5 and ψ the Riesz representative of the bounded functional $f \mapsto (u, Tf)$ on the closed subspace of divergence-free fields with zero boundary fluxes; so $\operatorname{div}\psi = 0$ and $F_i(\psi) = 0$. If $\operatorname{curl} w = c$, then c lies in that subspace and $w - Tc$ is curl free, so $(u, w) = (u, Tc) = (\psi, c)$ by Lemma 5.6. Taking test fields w gives $\operatorname{curl}\psi = u$; taking $w \in H^1(\Omega)^3$ gives $\psi \times n = 0$. Uniqueness is input 4. Conversely, inputs 5 and 7 give $u \cdot n = 0$ and $C_j(u) = 0$. □

Theorem 5.8 (The analytic inputs hold). [Lean: `VP.analyticInputs`] *For every admissible geometric datum the analytic inputs hold.*

Proof. Inputs 1–7 are Lemma 6.1, Theorem 8.3, Theorem 9.8, Theorem 9.9, Lemma 9.1, Lemma 6.4 and Lemma 6.5. □

Chapter 6

The cuts

Lemma 6.1 (The cuts are null)**.** [Lean: `VP.GraphPatch.graph_null, VP.cuts_null, VP.measure_cutDomain_eq`] *Every graph patch is Lebesgue null. Hence the union of the cuts is null, and the measures of Ω and Ω° restricted from Lebesgue measure coincide.*

Proof. Fubini's theorem in the coordinates of the patch: each vertical line meets the graph in one point. Each cut is covered by finitely many patches in which it is a graph. □

Lemma 6.2 (Unit-jump lifts)**.** [Lean: `VP.cutCutoff, VP.liftFn, VP.liftH1, VP.liftH1_unitJump, VP.liftQ, VP.hasWeakCurl_liftQ`] *For each cut Σ_j let χ_j be a smooth cutoff equal to 1 near $\overline{\Sigma_j}$ and supported in the union of its patches, and let M_j be the union of the minus sides. The lift $\ell_j = \chi_j \mathbf{1}_{M_j}$ belongs to $H^1(\Omega^\circ)$ and has unit jump across Σ_j. Its gradient $\mathbf{1}_{M_j}\nabla\chi_j$ is smooth on Ω, belongs to $H^1(\Omega)^3$ and is curl free on Ω.*

Proof. Off the cut, ℓ_j is locally χ_j or 0, hence smooth; near Σ_j, $\chi_j = 1$, so ℓ_j is locally constant on each side. The gradient vanishes near the cuts and is a gradient elsewhere. □

Lemma 6.3 (Gluing across the cuts)**.** [Lean: `VP.cut_glue`] *If $q \in H^1(\Omega^\circ)$ has unit jump across Σ_j, there is $r \in H^1(\Omega)$ with $\nabla r = \nabla q - \nabla \ell_j$, the gradients of q and ℓ_j being extended across the null cuts.*

Proof. $q - \ell_j$ has no jump across any cut. In each patch ball it is the restriction of an H^1 function by definition of the jump; away from the cuts it is H^1 already. A partition of unity glues the local weak gradients. □

Lemma 6.4 (Gradients of unit-jump functions are curl free)**.** [Lean: `VP.cutCurlFree_input`] *If $q \in H^1(\Omega^\circ)$ has unit jump across a cut, its extended gradient is curl free on Ω.*

Proof. It is $\nabla r + \nabla \ell_j$ with $r \in H^1(\Omega)$ (Lemma 6.3); both terms are curl free (Lemmas 5.2 and 6.2). □

Lemma 6.5 (Cut fluxes of curls)**.** [Lean: `VP.cutFlux_input`] *If $\psi \times n = 0$ for a field ψ with $\operatorname{curl}\psi = u$, and $u \cdot n = 0$, then $C_j(u) = 0$ for all j.*

Proof. For q with unit jump across Σ_j write $\nabla q = \nabla r + \nabla \ell_j$. Then $(u, \nabla r) = 0$ since $u \cdot n = 0$, and $(u, \nabla \ell_j) = (\psi, \operatorname{curl} \nabla \ell_j) = 0$ by $\psi \times n = 0$ tested with $\nabla \ell_j \in H^1(\Omega)^3$. □

Chapter 7

Analysis on $\mathbb{R}^3$

Lemma 7.1 (Mollification). [Lean: `VP.moll, VP.moll_contDiff, VP.eLpNorm_moll_le, VP.tendsto_moll`] *There is a smooth, nonnegative, radial kernel ρ_t of unit mass supported in the ball of radius t. For locally integrable f, $\rho_t * f$ is smooth; for $f \in L^2(\mathbb{R}^3)$, $\|\rho_t * f\|_{L^2} \le \|f\|_{L^2}$ and $\rho_t * f \to f$ in L^2 as $t \to 0$.*

Proof. The kernel is built from a smooth profile of $|y|^2$. Young's inequality gives the bound, and density of continuous compactly supported functions gives the convergence. □

Lemma 7.2 (Mollification commutes with weak derivatives). [Lean: `VP.WeakCurl, VP.WeakDiv, VP.gradTest_moll, VP.curlTest_moll`] *If f has the weak gradient g on an open set U and the closed ball of radius t around x lies in U, then $\nabla(\rho_t * f)(x) = (\rho_t * g)(x)$. The same holds for weak curls.*

Proof. Derivatives of $\rho_t * f$ are convolutions with derivatives of the kernel; at x the translated kernel is a test function on U, and the weak identity applies. □

Theorem 7.3 (Weyl's lemma for curl-free, divergence-free fields). [Lean: `VP.weyl`] *Let $U \subset \mathbb{R}^3$ be open and $H \in L^2(\mathbb{R}^3)^3$ weakly curl free and divergence free on U. Then H agrees almost everywhere on U with a smooth field that is classically curl free on U.*

Proof. At points whose τ-ball lies in U, the mollification $\rho_\tau * H$ does not depend on τ: the derivative of the radial kernel in τ is a divergence, whose pairing with H vanishes because H is curl free and divergence free. Letting $\tau \to 0$ identifies H with the smooth field $\rho_\tau * H$ locally, and Lemma 7.2 shows that this field is curl free. □

Lemma 7.4 (Translations). [Lean: `VP.tendsto_translate, VP.integral_inner_grad_eq_zero`] *Translation is continuous in $L^2(\mathbb{R}^3)$. If $f \in L^2(\mathbb{R}^3)$ has a weak gradient $g \in L^2$ and vanishes outside a compact subset of an open set U, then $\int_{\mathbb{R}^3} v \cdot g = 0$ for every $v \in L^2(\mathbb{R}^3)^3$ divergence free on U.*

Proof. Continuity of translation follows by density of continuous compactly supported functions. For the second statement, $\rho_t * f$ is a test function on U for small t, with gradient $\rho_t * g$, so $\int v \cdot (\rho_t * g) = 0$; let $t \to 0$. □

Chapter 8

Topology of the cut domain

Lemma 8.1 (Global primitives). [Lean: `VP.ClosedField, VP.exists_global_primitive`] *A smooth curl-free field on a simply connected open set $U \subset \mathbb{R}^3$ is the gradient of a smooth function on U.*

Proof. Local primitives on balls exist by the Poincaré lemma for convex sets. Pairs of a point and a value, topologized by the graphs of local primitives, form a covering space of U; since U is simply connected, the identity of U lifts to it, which is a global primitive. □

Lemma 8.2 (Primitives jump by constants across the cuts). [Lean: `VP.exists_cut_constant`] *Let h be a smooth curl-free field on Ω and P a primitive of h on Ω°. Along each cut Σ_j the jump of P across Σ_j is a single constant.*

Proof. Near a point of a cut, small boxes in the frame of the cut patch have path-connected sides. On each side P differs from a local primitive of h on the box by a constant, so the jump is locally constant along the cut, and the cut is connected. □

Theorem 8.3 (Orthogonality to harmonic fields). [Lean: `VP.harm_input`] *If $h \in L^2(\Omega)^3$ is curl free and divergence free, and $v \in L^2(\Omega)^3$ satisfies $v \cdot n = 0$ and $C_j(v) = 0$ for all j, then $(v, h)_{L^2(\Omega)^3} = 0$.*

Proof. By Theorem 7.3, h is smooth and curl free on Ω. On the simply connected cut domain it has a primitive P, which jumps by constants c_j across the cuts (Lemma 8.2). Then $p_0 = P - \sum_j c_j \ell_j$ is locally smooth on Ω with $\nabla p_0 = h - \sum_j c_j \nabla \ell_j$. The bounded truncations $n \arctan(p_0/n)$ lie in $H^1(\Omega)$, so $v \cdot n = 0$ and dominated convergence give $(v, \nabla p_0) = 0$, while $C_j(v) = 0$ gives $(v, \nabla \ell_j) = 0$. □

Chapter 9

The boundary

Lemma 9.1 (Zero tangential trace forces zero normal trace)**.** [Lean: `VP.ext0, VP.weakDiv_ext0_of_tangential, VP.orth_H1_of_weakDiv, VP.normalTrace_input`] *If $\psi \times n = 0$ for a field ψ with $\operatorname{curl}\psi = u$, then $u \cdot n = 0$.*

Proof. Testing $\psi \times n = 0$ with gradients of smooth compactly supported functions on $\mathbb{R}^3$ shows that the extension $\bar{u}$ of u by zero is divergence free on $\mathbb{R}^3$. For $q \in H^1(\Omega)$, a smooth partition of unity subordinate to Ω and to boundary graph patches localizes q. In a patch, $\bar{u}$ translated along the frame vanishes near the boundary graph, so it only meets a compactly supported cut-off of the localized function (Lemma 7.4); continuity of translation gives $\int_\Omega u \cdot \nabla q = 0$. □

Lemma 9.2 (Poincaré inequalities by segments)**.** [Lean: `VP.poincare_star, VP.PoincareSet, VP.PoincareSet.of_star, VP.PoincareSet.union`] *Let $S \subset \mathbb{R}^3$ be star shaped with respect to a set Z of positive finite measure, with $|w - x| \le d$ for $x \in S$, $w \in Z$. Then every smooth φ admits $c \in \mathbb{R}$ with*

$$\int_S |\varphi - c|^2 \le \frac{8d^2(|S| + |Z|)}{|Z|} \int_S |\nabla\varphi|^2.$$

Sets carrying such an inequality glue along overlaps of positive measure.

Proof. The fundamental theorem of calculus along the segments from $x \in S$ to $w \in Z$, Cauchy–Schwarz, and the affine changes of variables $x \mapsto (1-t)x + tw$ and $w \mapsto (1-t)x + tw$. □

Lemma 9.3 (Boundary charts and Lipschitz calculus)**.** [Lean: `VP.Chart, VP.exists_chart, VP.weakGrad_of_lipschitz, VP.lipschitz_cut`] *Every boundary point has arbitrarily small charts: a box V in frame coordinates in which Ω lies strictly above a globally Lipschitz graph and the exterior strictly below it. The exterior part $S = V \setminus \overline{\Omega}$ is star shaped with respect to its bottom slab, and the reflection across the graph is a Lipschitz, measure-preserving involution exchanging the two sides. A Lipschitz function on $\mathbb{R}^3$ has its almost-everywhere gradient as weak gradient, and a Lipschitz function vanishing on the zero set of a continuous function Φ stays Lipschitz after being cut off to $\{\Phi > 0\}$.*

Proof. The charts shrink the patches of the Lipschitz boundary. The weak gradient follows from Rademacher's theorem and integration by parts along lines; the cut-off statement from the intermediate value theorem. □

Lemma 9.4 (The reflection estimate)**.** [Lean: `VP.chart_estimate`] *Let u be divergence free on Ω, C a chart with exterior part S and Lipschitz constant K, ζ a smooth cutoff equal to 1 on the half box and supported in the box with $|\nabla\zeta| \le C_\zeta$, and F smooth with support in the half box. Then*

$$\Big|\int_\Omega u \cdot \nabla F\Big| \le \|u\|_{L^2(\Omega)^3}\big(C_\zeta\|F\|_{L^2(S)} + (3 + 2K)\|\nabla F\|_{L^2(S)}\big).$$

Proof. With R the reflection, $D = F - \zeta\, F \circ R$ is Lipschitz and vanishes on the graph. Cut off to the side of Ω and translated upward it becomes compactly supported in Ω, hence orthogonal to u through its gradient; continuity of translation gives $\int_\Omega u \cdot \nabla F = \int_\Omega u \cdot \nabla(\zeta\, F \circ R)$, which is bounded by the norms of F and ∇F on S because R exchanges the two sides and preserves measure. □

Theorem 9.5 (The key estimate). [Lean: `VP.ell`, `VP.key_estimate`] *Let $u \in L^2(\Omega)^3$ be divergence free with $F_i(u) = 0$ for all i. There are a bounded open set Θ disjoint from Ω and $K < \infty$ with $|\int_\Omega u \cdot \nabla\varphi| \le K\|\nabla\varphi\|_{L^2(\Theta)}$ for all smooth compactly supported φ on $\mathbb{R}^3$.*

Proof. Choose charts small enough to meet a single boundary component, and a partition of unity subordinate to Ω and finitely many charts. The interior piece of $\int_\Omega u \cdot \nabla\varphi$ vanishes; each chart piece is controlled by Lemma 9.4; on each boundary component a constant can be subtracted from φ because the fluxes vanish. The Poincaré inequalities of the exterior chart pieces chain along the connected boundary components (Lemma 9.2). □

Theorem 9.6 (Solenoidal extension). [Lean: `VP.exists_extension`] *A divergence-free $u \in L^2(\Omega)^3$ with $F_i(u) = 0$ for all i is the restriction to Ω of a compactly supported $U \in L^2(\mathbb{R}^3)^3$ that is weakly divergence free on $\mathbb{R}^3$.*

Proof. By Theorem 9.5, $\nabla\varphi|_\Theta \mapsto \int_\Omega u \cdot \nabla\varphi$ is a bounded functional on gradients in $L^2(\Theta)^3$. Extend it by Hahn–Banach and let w be its Riesz representative; then $U = u$ on Ω, $U = -w$ on Θ and $U = 0$ elsewhere is divergence free on $\mathbb{R}^3$. □

Lemma 9.7 (An explicit curl potential). [Lean: `VP.potOp`, `VP.weakCurl_potOp`] *Let $v \in L^2(\mathbb{R}^3)^3$ vanish outside the ball of radius ρ and be weakly divergence free on $\mathbb{R}^3$. If $\rho + \rho' \le R$, then*

$$(Av)(x) = \Big(0,\ \int_0^R v_3(x - te_1)\,\mathrm{d}t,\ -\int_0^R v_2(x - te_1)\,\mathrm{d}t\Big)$$

belongs to $L^2(\mathbb{R}^3)^3$ and has weak curl v on the ball of radius ρ'.

Proof. Transpose the line integrals (Fubini and translation invariance), so that the test field meets the adjoint line integrals; differentiate these under the integral sign. The fundamental theorem of calculus along e_1 produces v, because $v(x - Re_1) = 0$ for $|x| < \rho'$, and the remaining gradient term vanishes because $\operatorname{div} v = 0$. □

Theorem 9.8 (Divergence-free fields with zero fluxes are curls). [Lean: `VP.range_input`] *Every divergence-free $u \in L^2(\Omega)^3$ with $F_i(u) = 0$ for all i is a weak curl on Ω.*

Proof. Extend u by Theorem 9.6 and apply Lemma 9.7 on a ball containing Ω. □

Theorem 9.9 (Uniqueness of the normal potential). [Lean: `VP.weakCurl_ext0_of_tangential`, `VP.normalUnique_input`] *If $\psi \in L^2(\Omega)^3$ is divergence free, $\psi \times n = 0$ for $\operatorname{curl}\psi = 0$, and $F_i(\psi) = 0$ for all i, then $\psi = 0$.*

Proof. Let U be the solenoidal extension of ψ (Theorem 9.6) and $A = AU$ its potential (Lemma 9.7). The extension $\bar\psi$ of ψ by zero is curl free on $\mathbb{R}^3$ because $\psi \times n = 0$. Hence $\|\psi\|^2 = \int U \cdot \bar\psi = \lim_{t\to 0} \int U \cdot (\rho_t * \bar\psi) = \lim_{t\to 0} \int A \cdot \operatorname{curl}(\rho_t * \bar\psi) = 0$. □

Chapter 10

Correspondence with the paper

- **Formalized:** Theorem 3.12 (tangential vector potentials) and Theorem 3.17 (normal vector potentials), with the paper's hypotheses on Ω and on the cuts (Definition 3.3). Lean numbers the cuts $0, \dots, J-1$.
- **Function spaces.** Fields are classes in $L^2(\Omega)^3$, so uniqueness is uniqueness in $L^2(\Omega)^3$. In the paper ψ ranges over $X(\Omega) = H(\mathrm{curl}, \Omega) \cap H(\mathrm{div}, \Omega)$; this is the same, since $\mathrm{curl}\, \psi = u \in L^2(\Omega)^3$ and $\mathrm{div}\, \psi = 0$ already say that $\psi \in X(\Omega)$.
- **Boundary conditions and fluxes** are the paper's weak definitions by Green's formulas (Definitions 3.2 and 3.4). The flux conditions are imposed for every admissible cutoff and lift, which are shown to exist (Lemma 3.6); the paper shows that the fluxes do not depend on these choices.
- **Not part of the statements:** the additional sentences of Theorems 3.12 and 3.17 (linear dependence of ψ on u and the bound $\|\psi\|_X \leq C_\Omega \|u\|$). For the tangential potential, linearity and continuity in $L^2(\Omega)^3$ are proved in the library (Lemma 5.5).
- **Not formalized:** the other results of the paper (regularity and compactness of Section 2, the harmonic kernels, Helmholtz decompositions and the full Dirichlet potential of Section 3, and the finite-element and Stokes results of Section 4). Their mathematical proofs are in `blueprint/DetailedProofs.pdf`.